

PTC Uniform/Constant Analysis Coefficient

Engineering Conformance Specification

SCI-PTC-COEFFICIENT-UNIFORM v0.1-draft.1 / r0.4

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Abstract

Revised draft for scientific-owner review. This view is the complete proposed conformance specification: structural rules, independent fidelity facts, mixed-evidence cases, failure scopes, compact representation, publication lifecycle, requirements, predictions, and evidence mapping. Requirement verbs describe eventual obligations and do not authorize implementation or claim that any test has passed.

1 Program adherence and prior-work recovery

This implementation-blind revision follows only the released Stage A packet, the completed prior-work recovery synthesis, and the isolated scientific-owner review-1 supplement. It retains the frozen PTC coefficient taxonomy and lifecycle, the approved single-registry framework, the frozen MAP and JINC coefficient boundaries, the frozen NOI parent-binding limits, and the admitted VAL knowledge and named-use rules. Owner-requested recovery additionally cites only frozen SCI-PTC requirements 072, 090, 094, and 095 for atomic required outputs and ordinary-route rank provenance. No omitted parent material, implementation, test, audit, validation, or external scientific source was used.

The owner supports the fixed dimensionless-unity design, absence of a numerical parameter or default, separation from sample validity and uncertainty, separate MAP and JINC permissions, and the existing NOI limits in principle. The owner returned r0.2 for the targeted contract revisions rendered here and explicitly withheld complete contract approval and freeze. Every new family, profile, publication, permission, and boundary choice therefore remains an open proposal.

Revision r0.4 is a bounded consistency correction within scientific-owner round 1. It changes only the description of structural binding-record defects versus integrity of bytes in an already bound payload; it adds no source or scientific choice.

2 Scientific meaning of the uniform family

For every occurrence in its exact declared support, the PTC-owned coefficient is one. It is a transparent reference family: relative differences in a downstream contribution cannot originate from this coefficient. Other producer and consumer facts can still make contributions unequal or unusable. The constant is fixed at exact one because any positive common scale would be removed by mean-one normalization and would add a parameter with no surviving meaning.

Uniformity concerns only the positive PTC analysis/gridding coefficient. It does not assert equal signal quality, exposure, response, uncertainty, precision, significance, detector independence, or optimality. It does not restore an invalid sample, create support, or override a consumer gate. An occurrence authoritatively outside support has no coefficient under this family; absence is not numerical zero.

2.1 Handoff decisions and ownership

The proposed named-use profile asks whether one attempted handoff has the right selected family, exact parents and support, a faithful payload, and permission for one named consumer. The request to evaluate this profile is independent of the selected-family fact. Thus an explicit profile request with a known different selected family reaches UQ-01 and is ineligible; no explicit profile request creates no evaluation.

Payload fidelity is a producer-owned property shared by every handoff that uses the same immutable payload. It is composed from independently assessable scalar-value, unit, integrity, and support-membership facts. A decisive false in

any one fact is not hidden by an unrelated unknown. Contradictory claims about the same fact remain a conflict. A faithful payload is not made corrupt by consumer policy: a complete authoritative denial makes only that named use ineligible, while missing or conflicting permission evidence leaves that use decision unavailable absent another decisive false.

The scope follows the object that is defective. A malformed reference in one handoff request affects only that attempted request. A defect in a shared authoritative support, scalar publication, family generation, or source binding affects every request that references that object. A named-consumer permission result affects that consumer and does not determine the other consumer.

Classification follows the established identity. A corrupt authoritative identity or binding record can prevent structural establishment. Once the payload publication identity and bindings are valid, corruption, digest failure, or decoder failure in its bound bytes is a payload-integrity fact under $Q_{\text{integrity}}$, never structural merely because the bytes are corrupt.

Request, applicability, eligibility, and realization answer different questions. A requested and applicable profile can finish with an ineligible or decision-unavailable result; that completed negative or unresolved result is still a successfully realized scientific record. Failure to finalize the record instead leaves no completed evaluation to reinterpret. No exception, override, or advisory fact can bypass identity, payload fidelity, or the named-consumer permission needed for that use.

2.2 Consumer boundaries

For MAP, the proposed family supplies only the existing PTC coefficient facet; MAP retains its own admission, placement, exposure, coordinate, and numerical classification rules. For JINC, the PTC factor becomes one while JINC retains the signed spatial factor, normalization, conditioning, and support. MAP and JINC permissions are independent and remain proposed rather than granted.

A stored one never supplies missing selection, authority, provenance, or permission. The explicit PTC-disabled route remains RTC-terminal and creates no PTC coefficient or MAP route. The ordinary-route positive-rank limits are stated from their admitted frozen source; this contract infers no identity, zero-rank, or other route family. NOI still requires the exact lossless rational parent for the selected MAP and coefficient generations and gains no new route or design rule.

3 Registry entry and named-profile specification

3.1 Proposed family record

Facet	Proposed exact content
Registry and family	Registry <code>SCI-PTC:analysis_gridding_coefficients@draft-0.1</code> ; family <code>SCI-PTC:uniform_constant@draft-0.1</code> ; scientific owner SCI-PTC. The proposed actual human owner is Grant Wilson acting for SCI-PTC, pending UFC-Q-001. Both identities remain inactive drafts.
Numerical definition	Fixed raw scalar $c_{\star} = 1$, dimensionless unit 1, and named arithmetic-mean-to-one operator. No measured statistic, fitted factor, or configurable numerical parameter.
Logical index	Exact detector-time occurrence i , distinct from spatial consumer index p . Full stable occurrence identity is required.
PTC parent	One exact PTC transformed-product publication generation q bound to one application generation, immutable CAL parent, observation, and declared segment/scan/chunk bounds.
Support	Every and only exact occurrence in the output support of q ; finite, nonempty, and identity-unique. Observations and publication units are not pooled.
Publication unit	One independently complete, immutable q . Its declared unit kind is a segment, scan, or bounded chunk wholly within one observation and one PTC segment/scan identity. Handoff need not wait for other publication units in the observation.
Atomic completion	A q is complete only after all of its required components and links are durable. Partial output or failed required output is not labeled complete.
Representation	Exact scalar plus either a stored compact $\hat{R}_q$ checked against authoritative target R_q , or an enumerated member representation. A compact reference need not duplicate member identities. An enumerated form must prove every and only each target identity exactly once.

Facet	Proposed exact content
Shared evaluation	Family, scalar, unit, integrity, compatibility, and named permission facts may be evaluated once for their immutable scope and referenced by logical occurrence-level decisions. Per-occurrence materialization is not required.
Lifecycle	Family-rule version, application generation, PTC product generation q , coefficient realization g , payload publication, request, and profile evaluation are distinct immutable identities with one-way links.
Permissions	Separate proposed SCI-MAP and SCI-JINC grants. A denial is consumer-specific; missing or conflicting evidence is not a denial.
Uncertainty	The fixed coefficient introduces no sampling uncertainty. It supplies no value for signal, calibration, response, covariance, selection, representation, or performance uncertainty.
Prohibited meanings	Precision, inverse variance, sensitivity, significance, loading, scatter, validity, exposure, response, covariance, independence, optimality, default, support restoration, or missing-value replacement.

3.2 Profile identity, request, and applicability

The proposed profile is `SCI-PTC:uniform_coefficient_handoff@draft-0.1`, owned by SCI-PTC with the actual-owner field still open. Its named use is:

It explicitly binds the admitted SCI-PTC common named-use fragment at exact digest `c1fc8370007b65307769fb966c8523251695924aaff84f3e5b4c89b6d3380b8c`. That immutable common-fragment binding is part of the profile source identity. Its S_p, h_p binding names this exact contract revision and must be replaced by the promoted immutable source digest before activation.

For one explicit request naming (g, q, i, c) , decide whether the handoff selects `SCI-PTC:uniform_constant@draft-0.1`, binds the exact PTC and coefficient parents, uses a faithful exact-unity payload, and is authorized for named consumer $c \in \{\text{SCI-MAP}, \text{SCI-JINC}\}$.

A well-formed request names the profile, coefficient generation, PTC publication generation, occurrence, and consumer. This explicit named-profile request sets R = requested independently of the family-selection value. A known different selection is therefore a reachable UQ-01 false. If no named profile evaluation is requested, R = not requested and no eligibility question is evaluated. A malformed or ambiguous request reference cannot form that one question; its scope is the attempted request and it does not alter a shared generation. This includes one request that pairs q with the wrong otherwise-authoritative g .

For a well-formed request, applicability is true when q exists, its exact support is authoritatively finite and nonempty, and i belongs to it. Applicability is false when authoritative identity establishes that i is outside support. It is unknown when the target occurrence membership or required shared object cannot be established.

The aggregation status is $\Gamma_p = \text{none}$. This is an occurrence-level named-use profile. Reusing a shared scalar or fidelity result at its exact scope, and propagating a shared failure to all references, does not create an aggregate eligibility profile or an aggregate eligibility rule.

The lineage L_p one-way links the exact requested, effective-policy, observation-resolved, selected, coefficient-realized/published, and evaluation identities. Sharing a fact never reverses or collapses these lineage links.

The nonexceptionable shared-object gate requires active exact Registry, source, and owner bindings; complete application, CAL parent, observation, occurrence, q , g , family, selection-record identity, authoritative target-support reference R_q , payload-publication, target-broadcast, profile, and named consumer identities; the admitted VAL structural scope and influence bindings; and immutable provenance. A missing or conflicting selection-record identity is structural. After that identity is bound, its selection value is independently T/F/U/C under UQ-01.

A corrupt or misbound authoritative q -to- g identity/binding record, a missing or conflicting shared binding, or an authoritatively ambiguous population makes every question referencing that shared object unavailable. One attempted request that cites a wrong q, g pair remains request-local and does not contaminate the authoritative object.

3.3 Independent payload-fidelity facts

After structural establishment, UQ-05 consumes four facts about the same bound payload. Each fact has its own $\{\mathbf{T}, \mathbf{F}, \mathbf{U}, \mathbf{C}\}$ state and retained causes.

Fact	T	F	U and C
Q_{scalar}	Authoritative decoded scalar is exact one.	Authoritative decoded scalar is not exact one.	U : value evidence absent/unavailable. C : authoritative scalar claims for the same payload contradict or are ambiguous.
Q_{unit}	Authoritative decoded unit is dimensionless 1.	Authoritative decoded unit is different.	U : unit evidence absent/unavailable. C : same-payload unit claims contradict or are ambiguous.
$Q_{\text{integrity}}$	Declared decoder succeeds, publication digest and integrity checks pass, and the representation can encode the required value.	An authoritative check establishes digest/integrity failure, corruption, truncation, decoder-invalid bytes, or known unrepresentability, including corruption that makes the scalar unreadable.	U : required bytes, decoder, digest, or check unavailable, including unreadability without an authoritative cause. C : same-payload integrity claims contradict or are ambiguous.
Q_{member}	Stored compact $\hat{R}_g$ exactly matches the structurally valid target R_q , or enumerated form contains every and only each target identity exactly once.	Stored $\hat{R}_g$ names the wrong support/generation or mismatches target source/scope/digest, or a complete enumerated form omits, adds, or repeats a target member.	U : stored-reference comparison or complete enumeration evidence unavailable. C : same-payload stored-reference or membership claims contradict or are ambiguous.

A same-fact contradiction is one **C**; its negative branch is not also counted as an independent **F**. The exact UQ-05 aggregate is

$$Q_{05} = \begin{cases} \mathbf{F}, & \text{if any independent fidelity fact is } \mathbf{F}, \\ \mathbf{C}, & \text{if none is } \mathbf{F} \text{ and at least one is } \mathbf{C}, \\ \mathbf{U}, & \text{if none is } \mathbf{F}, \mathbf{C} \text{ and at least one is } \mathbf{U}, \\ \mathbf{T}, & \text{if all four facts are } \mathbf{T}. \end{cases}$$

Every component state and cause is retained. At profile level, one decisive restriction false yields ineligible even when another restriction is unknown or conflicting. With no false, any **U** or **C** yields decision unavailable; all required restrictions true yields eligible.

3.4 Restriction set

ID	Required predicate	False, missing, and conflict consequences
UQ-01	After the selection-record identity is structurally bound, its resolved value is exactly SCI-PTC: uniform_constant@draft-0.1 , with no numerical parameter or override.	Known different value is F and the requested profile is ineligible. Missing value in the bound record is U ; contradictory value claims are C .
UQ-02	An authoritative compatibility record permits g for its already bound q .	Explicit incompatibility after a valid shared q, g binding is F . A corrupt/misbound authoritative q -to- g identity/binding record is structural at shared scope; one attempted wrong pair is request-local. Missing/conflicting compatibility permission evidence is U/C .
UQ-03	The target support equals the complete output support of q , is finite and nonempty, and has unique occurrence identities.	Empty support creates no instance. Ambiguous/duplicate authoritative identity or conflicting shared support fails structural establishment.

ID	Required predicate	False, missing, and conflict consequences
UQ-04	The family rule fixes dimensionless $c_\star = 1$ and analytic $\mathcal{N}_{\text{mean1}}$ on the target support.	A known different constant, unit rule, operator, or population rule is F and another family/version; missing/conflicting rule evidence is U/C .
UQ-05	$Q_{\text{scalar}}, Q_{\text{unit}}, Q_{\text{integrity}}, Q_{\text{member}}$ compose by the exact rule above.	Any independent false is producer-invalid and makes every handoff using that payload ineligible. With no false, unknown/conflict makes the decision unavailable.
UQ-06	The complete authoritative permission record explicitly authorizes the named consumer c .	A complete record that denies or does not authorize c is F for that consumer only. Missing/unresolved evidence is U ; contradictory evidence is C . Payload fidelity and the other consumer are unchanged.
UQ-07	Pre-existing parent/lineage facts and the proposed write-once construction satisfy the staged lifecycle rule in the lifecycle section.	A known pre-finalization lineage or nonmutation violation is F . Missing/conflicting lifecycle evidence is U/C . Finalization failure sets profile realization failed rather than inventing a completed UQ-07 result.

All restrictions and structural invariants are nonexceptionable. Sample validity, retention, MAP/JINC admission, response, covariance, and diagnostics remain separate facts. Response and uncertainty availability are advisory and neutral for this coefficient-fidelity profile.

4 Formal consumer-boundary proposal

The proposed MAP boundary successor is `SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4`. It changes only the coefficient-slot binding. MAP may retrieve exact $\gamma_i = \omega_i = 1$ only after the family record explicitly grants SCI-MAP permission and the named profile evaluation is requested, applicable, eligible, and realized. MAP retains every existing admission, placement, exposure, response, coordinate, support, arithmetic, failure, and finite-strictly-positive classification rule.

The proposed JINC boundary successor is `SCI-PTC_TO_SCI-JINC_BOUNDARY v0.2-draft.1/r0.4`. It changes only the PTC coefficient binding and requires separately explicit SCI-JINC permission. JINC retains ownership of its spatial factor κ_{ip} , its sign, normalization, conditioning, support, gates, and bundle semantics. No response, covariance, uncertainty, or new bundle role is created.

Neither consumer may infer the family from a value, unit, label, another consumer, or missing selection. There is no unity, alternate-family, neighboring-product, direct-CAL, no-op-PTC, reconstruction, or generation-repair fallback. The NOI design law and exact rational-parent requirement remain unchanged.

5 Mathematical definition, support, and publication

5.1 Distinct identities and indices

Let a identify one application generation. Let q identify one exact, immutable, completed PTC transformed-product publication generation bound to a , its CAL parent, and one observation o . Let g identify one coefficient realization bound to q . The logical coefficient index i is an exact detector-time occurrence with stable identity ι_i . The symbol p is reserved for a consumer-owned spatial/map index; it never denotes a PTC product.

Each q declares exactly one publication-unit tag and bounds: a complete segment, a complete scan, or a bounded chunk wholly within one named segment/scan and observation. A unit never crosses observation, application, CAL-parent, segment/scan, or PTC-product-generation identity. The tagged unit size is explicit in q ; this contract supplies no hidden size default.

5.2 Exact support

The coefficient support and normalization population are

$$\mathcal{D}_{g,q} = \mathcal{S}_{g,q}^{\text{coef}} = \mathcal{S}_q^{\text{PTC,out}}. \quad (1)$$

They contain every and only exact occurrence in the completed output support of q . The population is finite, nonempty, and identity-unique. It may span all detectors, groups, arrays, networks, and times present in that publication unit; none silently defines a separate population. A singleton is valid. An empty population has no family instance. Observations and publication units are never pooled.

A required q is labeled complete only after all required components and links for that unit are durable, as required by frozen SCI-PTC v0.1/r0.5 requirement 072. This preserves the inherited atomic-output rule while allowing each completed segment, scan, or bounded chunk to be published and handed off independently. No coefficient handoff waits for unrelated publication units or for the whole observation to finish.

5.3 Uniform construction

For finite nonempty $\mathcal{D}_{g,q}$,

$$c_\star \equiv 1, \quad [c_\star] = 1, \quad (2)$$

$$\tilde{\omega}_i = c_\star, \quad i \in \mathcal{D}_{g,q}, \quad (3)$$

$$m_g = \frac{1}{|\mathcal{D}_{g,q}|} \sum_{j \in \mathcal{D}_{g,q}} \tilde{\omega}_j = 1, \quad (4)$$

$$\omega_i = \mathcal{N}_{\text{mean1}}(\tilde{\omega}_i; \mathcal{D}_{g,q}) \equiv \frac{\tilde{\omega}_i}{m_g} = 1. \quad (5)$$

The mean is a mathematical identity, not an instruction to execute a population-wide floating-point reduction of stored ones. A conforming realization may return exact unity analytically after the finite, nonempty support fact is established. Changing the normalization rule changes the family definition even though this family's result is one.

The family accepts no numerical parameter. A request carrying another constant, including another positive constant whose scale would cancel, is not an instance of this family and cannot be clipped, ignored, or normalized into acceptance.

5.4 Compact exact association

Let R_q be the immutable authoritative target-support reference in Equation 1. Its identity, source, generation, scope, and digest are structurally bound to q , and

$$\text{resolve}(R_q) = \{\iota_i : i \in \mathcal{D}_{g,q}\}. \quad (6)$$

A compact scalar publication carries a distinct stored support reference $\hat{R}_g$. Its membership fact is true only when the independently checked $\hat{R}_g$ exactly identifies R_q , including matching source, generation, scope, and digest. Equation 6 defines the target association; $\hat{R}_g$ is evidence that the payload realizes it without a copied member list. If a representation instead enumerates members, its complete stored manifest M_g must satisfy

$$M_g = \{(\iota_i, 1) : i \in \mathcal{D}_{g,q}\} \quad \text{with each identity exactly once.} \quad (7)$$

A missing or conflicting authoritative R_q prevents structural establishment. After R_q is valid, a wrong or mismatched stored $\hat{R}_g$, or an omission, addition, or repetition in M_g , is a payload-membership false. An unavailable stored reference or enumeration check is unknown; contradictory claims about that same stored fact are conflict. Row order, shape, cardinality, detector label, coordinate equality, or container position cannot replace exact identity.

Family identity, scalar value, unit, integrity, q -to- g compatibility, and named permission are immutable shared facts. They may be evaluated once for their declared scope. Each occurrence-level handoff remains logically associated with those shared evaluations through exact references plus its own $i \in \text{resolve}(R_q)$ and request identity. Neither the coefficient array nor a separate full profile record must be materialized for every occurrence.

5.5 Consumer consequences

At the JINC boundary, p is JINC's spatial index and

$$w_{ip} = \kappa_{ip}\omega_i = \kappa_{ip}. \quad (8)$$

This equality transfers no ownership of κ_{ip} , its sign, or JINC support.

For the already selected NOI design on an exact MAP-admitted contribution population, p is the MAP spatial index, C_p is MAP's occurrence set for that spatial index, and

$$a_{pi} = G_{pi}\gamma_i = G_{pi}, \quad B_d = \sum_p \sum_{\substack{i \in C_p \\ \text{detector}(i)=d}} a_{pi}. \quad (9)$$

This does not define G_{pi} , change MAP arithmetic, or alter the NOI design law.

5.6 Representation and publication conformance

Form or stage	Required evidence	Forbidden inference
Compact support reference	Structurally valid target R_q , stored $\hat{R}_g$, and an independent exact identity/source/generation/scope/digest match between them.	Cardinality, order, shape, or a numerical one cannot substitute; a wrong stored reference is payload false, not a structural re-definition of R_q .
Enumerated representation	Complete M_g , every and only target identity, each exactly once, plus scalar/unit/integrity facts.	Deduplication, truncation, inferred members, or acceptance from matching length is forbidden.
Shared fact evaluation	Immutable fact result, exact declared scope, source and input identities, retained T/F/U/C causes, and a stable reference from each logical handoff.	Sharing cannot widen scope or erase an occurrence-specific membership/request result.
Publication completion	All required components and links for that q are durable before complete status; required failure propagates.	A partial chunk cannot be labeled complete, and incomplete status cannot be repaired by unity.
Incremental handoff	A completed q , its exact support and payload, and a complete profile decision for the requested use.	Completion of other segments, scans, chunks, or the whole observation is not a prerequisite.
Normalization evaluation	Finite nonempty support plus the exact registered constant and operator establishes $m_g = 1$ analytically.	A literal population reduction or per-member stored coefficient is not required.

6 Scientific assumptions, uncertainty, and route limits

The family assumes only a completed, exact PTC publication generation q with identifiable finite nonempty output support. It assumes no statistical distribution, stationarity, independence, equal detector response, equal exposure, residual model, sensitivity model, or covariance model. The coefficient is defined rather than estimated and ranks no data quality.

The exact policy constant introduces no coefficient sampling uncertainty. This does not assign zero to signal, calibration, response, covariance, support-selection, family-selection, representation, or performance uncertainty. Unknown remains unknown, and absent covariance does not mean independence.

6.1 Ordinary-route and rank provenance

Frozen SCI-PTC v0.1/r0.5 requirement 090 defines the admitted first ordinary PTC route: the parent q is an explicitly requested calibrated- x , configured-rank, group-local PCA/SVD operation with exact CAL parent, segment, output identity, lifecycle, and failure state. Requirement 094 requires an integer rank at least one for each group, or one explicitly

common positive rank applied to named groups; every group verifies realizability on its own support and degeneracy state. Requirement 095 says rank zero, noninteger rank, or an unrealizable positive rank fails closed with exact group and cause. It does not become centering-only output, get clipped or substituted, become PTC-disabled, select RTC-terminal export, or produce a map.

Consequently, this coefficient can be instantiated on the ordinary route only from a completed q produced after those rank obligations pass. The PTC-disabled route remains separately RTC-terminal and produces no PTC coefficient. These inherited statements provide no authority for an identity PTC route, a zero-rank route family, or any other route.

6.2 Family rules, realizations, and immutable change

A family-rule version defines the constant, unit, normalization operator, support rule, representation permissions, and profile semantics. Changing one of those scientific rules creates the appropriate versioned family, profile, permission, or boundary successor.

Applying unchanged rules to a different application generation, observation, publication unit, or exact support population creates new immutable q and g realizations under the same family version. It does not version the family. A published realization is never rewritten: a corrected or changed actual support produces a new realization and preserves the earlier record. Numerical equality alone does not authorize cross-generation borrowing.

7 Scientific choices still awaiting owner approval

Owner review round 1 supports the basic uniform design in principle and withholds approval of the complete contract. The owner still must adopt, revise, or reject:

- the registered exact-unity family and analytic mean-one convention;
- the exact publication-unit support rule, compact support reference, shared-fact association, and incremental handoff proposal;
- the named-profile identity, independent request rule, four fidelity facts, mixed-evidence composition, nonexceptionability, and failure scopes;
- the separate MAP and JINC grants and their r0.4 boundary successors;
- immutable realization, replay, supersession, uncertainty, and Registry source-binding rules; and
- the proposed actual-owner field, Grant Wilson acting for SCI-PTC.

The first five groups are complete author proposals rather than missing definitions. The final item remains precise question UFC-Q-001 and blocks active Registry/profile binding. No implementation conformity, observational performance, production readiness, Registry activation, default, or complete scientific freeze follows from this revision.

8 Decision semantics and failure scopes

8.1 Knowledge states and four independent axes

Every fact uses $\{\mathbf{T}, \mathbf{F}, \mathbf{U}, \mathbf{C}\}$. Silence is unknown. Contradictory authoritative claims about the same fact remain $\mathbf{C}$. Across independent facts, a decisive $\mathbf{F}$ dominates unrelated unknown or conflict for eligibility while every cause remains recorded. With no false, any required unknown or conflict yields decision unavailable; all required facts true yields eligible.

The named profile carries independent axes:

- R : requested or not requested, set by the explicit named-profile request rather than by the selected-family value;
- A : applicable, inapplicable, or applicability unknown;
- E : eligible, ineligible, decision unavailable, or uninstantiated where no question is evaluated; and
- Z : realized, incomplete, failed, or not produced for the profile evaluation artifact, separate from payload availability.

A completed false predicate produces a successfully realized negative evaluation, $Z = \text{realized}$ and $E = \text{ineligible}$. A completed record of missing authority may likewise have $Z = \text{realized}$ and $E = \text{decision unavailable}$. $Z = \text{failed}$ is reserved for failure to complete or publish the profile-evaluation artifact.

8.2 Failure-scope table

Established condition	Required result	Scope
Named request and evaluation identity are recordable, but its occurrence, consumer, selection-record, or q, g reference is malformed, missing, ambiguous, or pairs the wrong otherwise-authoritative objects	$R = \text{requested}$, $A = \text{unknown}$, $E = \text{decision unavailable}$ in a completed request-local record; no shared object changes	That attempted request only
Request lacks enough identity to create a durable named-profile record	No eligibility assertion; evaluation artifact is failed or not produced	That attempted request only
An authoritative identity/binding record for Registry/source/owner, selection, application/CAL, q -to- g , target R_q , payload publication, or target broadcast is missing, conflicting, corrupt, or misbound; or the authoritative population identity is ambiguous	Applicability unknown or decision unavailable; no UQ-05 evaluation or fallback	Every request referencing the defective shared object
Known empty output support of q	No family instance, coefficient realization, or eligibility claim	That publication unit
Explicit named profile request with known different selected family or known incompatible q -to- g permission	UQ-01 or UQ-02 is F ; ineligible without reinterpreting the payload	Every request sharing that exact selection or compatibility fact
Any independent payload-fidelity fact is F	UQ-05 is F ; payload producer-invalid; completed handoff evaluations ineligible	Every request using that payload publication
No fidelity false, but a required fidelity fact is U or C	UQ-05 remains unknown/conflicting; decision unavailable unless another restriction is false	Every request using that payload publication
Complete permission record denies or does not authorize named consumer c	UQ-06 is F ; that named use is ineligible; payload and other consumer unchanged	Consumer c under that permission record
Permission evidence for c is missing/unresolved or contradictory	UQ-06 is U or C ; decision unavailable absent another false	Consumer c under that permission record
Occurrence is authoritatively outside q 's support	Inapplicable, $E = \emptyset_E$, and no coefficient for that occurrence	That occurrence request
Consumer rejects an otherwise faithful value under its own numerical rule	Preserve PTC profile; consumer excludes or fails under its rule	That consumer contribution
Advisory response, uncertainty, or unrelated diagnostic is missing/conflicting	Preserve exact state and cause; neutral for coefficient fidelity	No coefficient-profile failure

In the structural row, “corrupt” modifies an authoritative identity or binding record, not bytes of an already bound payload. After valid publication identity and binding, authoritative payload-byte corruption, digest failure, or decoder-invalid bytes gives $Q_{\text{integrity}} = \text{F}$; unavailable or unexplained integrity evidence gives **U**; and contradictory integrity evidence gives **C**. Payload bytes never become structural merely because they are corrupt.

8.3 Noncircular UQ-07 construction and finalization

UQ-07 is staged and never requires an already completed evaluation of itself.

1. Construction allocates the evaluation identity and snapshots the pre-existing request, Registry, parent, lineage, support, payload, permission, and UQ-01–UQ-06 fact references.

2. Before sealing, UQ-07 evaluates only those pre-existing lineage, nonmutation, scope, and write-once-target facts. A known violation is **F**; missing/conflicting evidence is **U/C**.
3. Finalization durably writes the complete result and its retained causes, then the lifecycle envelope records $Z =$ realized. If durable finalization fails, the envelope records failed/incomplete as available; no completed UQ-07 value or eligibility assertion is invented.

Later discovery never rewrites the sealed record. It produces a successor evaluation or realization with explicit supersession and preserved cause.

9 Evidence status and compact owner-decision register

Algebraic equality follows from the exact definition. Implementation conformity, representation and response fidelity, observational performance, and production readiness are separate evidence layers and have not been established.

This register is checked against the companion owner ledger.

Recovered and deferred states

IDs	State	Disposition
UFC-REC-001–008	decided	Admitted PTC, Registry, MAP, JINC, NOI, VAL, PTC-disabled, atomic-output, and ordinary-rank facts are recovered constraints; no claim is blocked.
UFC-DEF-001–005	deferred	Other coefficient families, defaults, NOI profile successors, performance criteria, and a general cross-generation alias remain outside this draft.

Open author proposals

ID	State	Exact blocked claim or output
UFC-OD-001	open	Registered exact-unity numerical family and payload.
UFC-OD-002	open	Exact logical index and compact occurrence-association forms.
UFC-OD-003	open	Publication-unit support and incremental handoff rule.
UFC-OD-004	open	Analytic mean-one operator and empty/singleton behavior.
UFC-OD-005	open	No-parameter and no-default selection rule.
UFC-OD-006	open	Named-profile identity, independent request, object, and applicability.
UFC-OD-007	open	Fidelity-fact composition, restrictions, nonexceptionability, permission distinctions, and failure scopes.
UFC-OD-008	open	SCI-MAP permission and MAP use of this family.
UFC-OD-009	open	SCI-JINC permission and JINC use of this family.
UFC-OD-010	open	Immutable MAP and JINC r0.4 successor-boundary bindings.
UFC-OD-011	open	Exact actual-owner field; the mathematical draft is not blocked.
UFC-OD-012	open	Realization, replay, finalization, family-version, and cross-generation rules.
UFC-OD-013	open	Coefficient uncertainty meaning.
UFC-OD-014	open	Active coefficient Registry identity and source binding.

Precise open question

ID	State	Exact question and blocked output
UFC-Q-001	open	Is the actual scientific owner Grant Wilson acting for SCI-PTC, as proposed, or another named authority? This blocks active Registry/profile binding only.

10 Normative proposed engineering requirements

Every requirement is an open proposal pending scientific-owner adoption.

ID	Requirement
UFC-REQ-001	The family shall retain exact identity <code>SCI-PTC:uniform_constant@draft-0.1</code> , owner SCI-PTC, and inactive draft status until exact owner-approved source and Registry bindings exist.
UFC-REQ-002	The coefficient index shall be exact detector-time occurrence i , distinct from spatial index p . A scalar shall bind occurrences by either stored compact reference $\widehat{R}_g$ independently matched to authoritative target R_q , or complete enumerated manifest M_g ; order, shape, label, or cardinality shall not substitute.
UFC-REQ-003	Each coefficient realization shall bind one independently complete PTC publication generation q , tagged as a segment, scan, or bounded within-scan/segment chunk in one observation. Its support shall equal the exact output support of q ; completion of unrelated units or the whole observation shall not gate handoff.
UFC-REQ-004	On finite nonempty support, the raw constant shall be exact dimensionless one and arithmetic-mean-to-one normalization shall be evaluated analytically as one. A literal population-wide reduction or stored per-occurrence array shall not be required.
UFC-REQ-005	The family shall accept no numerical parameter. An altered constant, unit rule, operator, or population rule shall be another family/version and shall not be ignored, clipped, or normalized into acceptance.
UFC-REQ-006	A singleton target shall be valid. Empty support shall create no instance. Duplicate, ambiguous, or conflicting identity in the authoritative target shall fail structural establishment and shall not be deduplicated or repaired.
UFC-REQ-007	An occurrence authoritatively outside q 's support shall have no coefficient under this family; absence shall remain distinct from numerical zero.
UFC-REQ-008	Every q and g shall bind exact application generation, CAL parent, observation, segment/scan/chunk bounds, ordinary-route rank provenance where applicable, selection lineage, support, source, and durable required components/links without mutating a parent or earlier realization.
UFC-REQ-009	Immutable family, scalar, unit, integrity, compatibility, and permission evaluations may be shared once at their exact scope. Every occurrence-level handoff shall retain a logical association to those results and its own membership/request fact; redundant per-occurrence materialization shall not be required.
UFC-REQ-010	SCI-MAP and SCI-JINC permission facts shall be separate. A denial, absence, unknown, or conflict for one shall not change payload fidelity or determine the other.
UFC-REQ-011	An explicit request for <code>SCI-PTC:uniform_coefficient_handoff@draft-0.1</code> shall set the request axis independently of family selection. After the selection-record identity is structurally bound, a known different value shall reach UQ-01 and be false, missing value shall be unknown, and conflicting value claims shall conflict. No named-profile request shall create no evaluation.
UFC-REQ-012	The complete profile shall bind exact use, owner, source, (g, q, i, c) object, applicability, shared-object gate, restrictions, fact-level truth rules, no-exception rule, advisory roles, compatibility, supersession, scopes, aggregation status, and lifecycle.
UFC-REQ-013	Structural identity, scalar value, unit, integrity, stored membership, payload availability, producer validity, named permission, sample validity, consumer admission, numerical classification, and contribution shall remain distinct facts.
UFC-REQ-014	$Q_{\text{scalar}}, Q_{\text{unit}}, Q_{\text{integrity}}, Q_{\text{member}}$ shall be independently T/F/U/C. Any independent false shall make UQ-05 false; otherwise conflict shall remain conflict, otherwise unknown shall remain unknown, and only all true shall yield true. Contradiction within one fact shall remain one conflict rather than a selected false.
UFC-REQ-015	Requested, applicability, eligibility, and profile-artifact realization shall be independent axes. A completed false predicate shall yield a realized ineligible evaluation; a completed unresolved evaluation may be realized and decision unavailable; failed realization shall mean the profile artifact itself failed.
UFC-REQ-016	Every profile restriction and structural invariant shall be nonexceptionable. No VAL, consumer, mode, or other-use exception shall alter the result.

ID	Requirement
UFC-REQ-017	Failure scope shall follow the defective object. A wrong q, g pair or other malformed reference in one request shall remain local; a corrupt/misbound authoritative q -to- g or other shared identity/binding record shall affect every reference to that record. After valid payload-publication binding, corrupt bytes, digest failure, and decoder failure shall remain $Q_{\text{integrity}}$ rather than structural; payload false shall affect that payload's handoffs. Permission state shall affect only its named consumer. Every cause and independent fact state shall be retained.
UFC-REQ-018	Response and uncertainty availability shall be advisory and neutral for coefficient fidelity. Exact upstream states shall be preserved, and neither identity response nor zero uncertainty shall be inferred.
UFC-REQ-019	The coefficient shall be nonprecision and shall not mean inverse variance, sensitivity, significance, fitted loading, scatter, sample validity, exposure, response, covariance, independence, optimality, or a policy default.
UFC-REQ-020	The MAP successor shall change only its coefficient binding, retain all MAP-owned gates and arithmetic, and require MAP's independent finite-strictly-positive classification before use.
UFC-REQ-021	The JINC successor shall change only its PTC-coefficient binding, retain JINC ownership of spatial index p , κ_{ip} , sign, arithmetic, gates, and support, and create no new response, covariance, uncertainty, or bundle role.
UFC-REQ-022	NOI compatibility shall require the frozen exact lossless rational-parent record for the exact MAP and coefficient generations. Missing binding shall leave NOI unavailable and shall not change its design law.
UFC-REQ-023	No family or publication-size default shall be inferred. Missing selection or authority shall produce no unity or alternate-family fallback. PTC-disabled shall remain RTC-terminal. On the admitted first ordinary route, rank zero, noninteger rank, or unrealizable positive rank shall fail closed and shall not select another route or produce a coefficient. No claim is made about unapproved identity or zero-rank route families.
UFC-REQ-024	A change to a scientific family/profile/permission/boundary rule shall create the corresponding versioned successor. New application, product, support population, payload, request, or evaluation under unchanged rules shall create a new immutable realization, not a new family version; no earlier record shall be rewritten.
UFC-REQ-025	Algebraic conformance, implementation conformity, representation/response fidelity, observational performance, and production readiness shall remain separate evidence claims. No implementation or validation result shall be inferred from this specification.

11 Worked decision cases

These are proposed discriminating cases, not observations of an implementation.

ID	Established evidence	Required result
UFC-MIX-001	Target established; $Q_{\text{scalar}} = \mathbf{F}$ from exact stored value two; $Q_{\text{member}} = \mathbf{U}$; unit and integrity true.	UQ-05 is $\mathbf{F}$. Payload is producer-invalid and all of its handoffs are ineligible; membership unknown is retained.
UFC-MIX-002	Target established; complete enumeration omits a target member, so $Q_{\text{member}} = \mathbf{F}$; integrity check is unavailable; scalar and unit true.	UQ-05 is $\mathbf{F}$; integrity unknown is retained and does not hide the membership violation.
UFC-MIX-003	Two authoritative scalar claims for the same payload contradict; all other fidelity facts are true.	$Q_{\text{scalar}} = \mathbf{C}$, UQ-05 is $\mathbf{C}$, and eligibility is decision unavailable. Neither scalar claim is selected.
UFC-MIX-004	The same scalar conflict exists, and the independently assessed enumeration repeats one target member.	$Q_{\text{scalar}} = \mathbf{C}$, $Q_{\text{member}} = \mathbf{F}$, and UQ-05 is $\mathbf{F}$. Eligibility is ineligible while the scalar conflict remains recorded.
UFC-MIX-005	Payload fidelity is true; a complete permission record does not authorize MAP; JINC permission is assessed separately.	MAP UQ-06 is $\mathbf{F}$ and MAP use is ineligible. Payload remains faithful; no JINC result is inferred.

ID	Established evidence	Required result
UFC-MIX-006	Payload fidelity is true; MAP permission evidence is missing or unresolved, with no other false.	MAP UQ-06 is U ; MAP decision is unavailable, not denied. Payload and JINC remain unchanged.
UFC-MIX-007	One named-profile request has a malformed occurrence reference or pairs the wrong otherwise-authoritative q, g ; the shared objects are valid.	Only that request is unavailable or unrecordable according to its preserved identity state. Other well-formed requests remain assessable.
UFC-MIX-008	The authoritative shared support R_q has ambiguous occurrence identity, or the shared q -to- g identity/binding record is corrupt/misbound.	Structural establishment fails for every request referencing that shared object; UQ-05 is not evaluated.
UFC-MIX-009	Compact form stores exact one and $\hat{R}_g$, which an independent check matches exactly to valid target R_q , including source, generation, scope, and digest.	$Q_{\text{member}} = \mathbf{T}$ without a duplicated member manifest; logical occurrence handoffs may reference the shared result. A wrong stored $\hat{R}_g$ would be F , not a structural change to R_q .
UFC-MIX-010	Enumerated form has the right length but repeats one identity and omits another.	$Q_{\text{member}} = \mathbf{F}$; length cannot repair identity mismatch, and the shared payload is producer-invalid.

12 Predictions and discriminating edge cases

ID	Prediction or edge case
UFC-PRED-001	For every finite nonempty exact support and every member, $\tilde{\omega}_i = \omega_i = 1$; all pairwise coefficient ratios and the arithmetic mean are exactly one.
UFC-PRED-002	A publication support containing exactly one occurrence is valid and produces one exact coefficient.
UFC-PRED-003	Empty support produces no mean, coefficient realization, payload, or eligibility claim.
UFC-PRED-004	Completed segment, scan, or bounded-chunk publication units can be handed off independently without an observation-wide barrier. Different actual units create new q, g realizations under the same family rule.
UFC-PRED-005	Repetition from the same exact inputs is numerically identical, yet each q, g and evaluation retains immutable identity; unbound cross-generation substitution is forbidden.
UFC-PRED-006	When the named profile is explicitly requested, a selected constant override or different family reaches UQ-01 and is ineligible rather than silently ignored.
UFC-PRED-007	A false scalar, unit, integrity, or membership fact makes UQ-05 false even with unrelated unknown/conflict. The payload is producer-invalid, while the completed negative profile artifact is realized and ineligible.
UFC-PRED-008	With no fidelity false, absent evidence or unexplained unreadability leaves UQ-05 unknown; same-fact contradiction remains conflict. The decision is unavailable without unity fallback.
UFC-PRED-009	A stored compact $\hat{R}_g$ exactly matched to target R_q establishes membership without a duplicated list. A wrong stored reference is payload false; missing/conflicting target or corrupt shared identity/binding record has shared structural scope; a wrong pair in one request remains request-local. After valid binding, bound-payload byte corruption remains $Q_{\text{integrity}} = \mathbf{F}$.
UFC-PRED-010	An invalid signal occurrence can still have a faithful coefficient one. It contributes only if independent producer and consumer gates admit the signal and occurrence.
UFC-PRED-011	MAP consumption gives $\gamma_i = 1$ and removes only variation attributable to the PTC coefficient; other MAP factors and gates remain.
UFC-PRED-012	JINC consumption gives $w_{ip} = \kappa_{ip}$, including JINC's sign; final JINC weights need not be positive or uniform.
UFC-PRED-013	An authoritative MAP denial affects only MAP; missing or conflicting MAP permission is decision unavailable absent another false. Neither state determines JINC or payload fidelity.
UFC-PRED-014	With no named-profile request there is no evaluation. With no selected family and no authorized default, neither consumer may infer this family from a stored one, a label, or another route.

ID	Prediction or edge case
UFC-PRED-015	On an otherwise valid exact MAP host route, $a_{pi} = G_{pi}$; missing lossless rational identity for that exact parent still leaves NOI design resolution unavailable.
UFC-PRED-016	On the admitted first ordinary route, missing valid positive integer rank, noninteger/zero rank, or unrealizable positive rank prevents a completed q , coefficient, and downstream route and does not become PTC-disabled or another route. PTC-disabled separately produces no PTC coefficient. No identity or zero-rank route family is inferred.

13 Evidence mapping after owner adoption

Requirement group	Required evidence	Discriminating coverage
UFC-REQ-001–010	Source/Registry identity, algebraic, publication, representation, and life-cycle conformance	Exact one and singleton/empty support; analytic normalization; compact and enumerated forms; independent unit completion; shared fact references; immutable replay.
UFC-REQ-011–018	Profile, truth-algebra, request, scope, and four-axis conformance	UFC-MIX-001–010; named request with wrong family; permission denial versus unknown/conflict; successful negative evaluation versus failed artifact; advisory facts neutral.
UFC-REQ-019–024	Meaning, consumer boundary, rank provenance, fallback, NOI, and change-control conformance	No precision/default meaning; MAP/JINC permission isolation; ordinary-rank failures; PTC-disabled separation; exact NOI parent; family version versus realization.
UFC-REQ-025	Evidence-layer reporting	Algebraic, implementation, representation/response, observational, and readiness claims reported separately with exact identities.

Conformance evidence must bind the exact approved source, Registry record, application, q , g , support reference or enumeration, payload, request, consumer, and evaluation identities. It must exercise all T/F/U/C compositions and preserve every cause. Successful scan/segment/chunk evidence must show that an independently complete unit can be handed off before unrelated observation units without labeling partial required output complete.

Scientific validation remains separate and future. Comparative performance or suitability would require owner-approved populations, questions, metrics, uncertainty treatment, and criteria. No larger observational program is invented for the algebraic identity $1 = 1$, and no implementation has been inspected.